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Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses $v = u + at$ with $v = 0$ for the vertical motion Condone cos or sign error	AO3.4	M1	$0 = u \sin 35 - 9.81 \times 1.5$ $u = 25.7$ m s^{-1}
	Obtains correct equation	AO1.1b	A1	
	Obtains correct u to 3 significant figures **CAO**	AO1.1b	A1	
(b)	Uses $s = ut + \frac{1}{2}at^2$ with $s = -10$ and their u for vertical motion Condone cos or sign error	AO3.4	M1	$-10 = (25.7\sin 35)t - \frac{1}{2} \times 9.81t^2$ $t = 3.571$ Time in flight is 3.57 seconds
	Obtains correct equation	AO1.1b	A1F	
	Obtains correct time of flight with units AWRT 3.6 **CAO**	AO3.2a	A1	
				Total 6 marks

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Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$
	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their'	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$

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	derived values for U and V			
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067\ldots$
	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3\text{ m s}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part **(a)**	AO3.4	A1F	16 m s $^{-1}$
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary… air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q3.

(a) $(V_H =) \dfrac{38.4}{2.4} = 16\text{ m s}^{-1}$

M1: Horizontal range divided by time.

A1: Correct speed.

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M1A1

2

(b) $3 = V_v \times 2.4 - \dfrac{1}{2} \times 9.8 \times 2.4^2$

M1: Equation to find the vertical component, with $s = \pm 3$, $t = 2.4$ and $a = \pm g$ or ±9.8 or ±9.81.

A1: Correct equation with g or 9.8 or ±9.81.

M1A1

$V_v = \dfrac{3 + 28.224}{2.4} = 13.01$

A1: Correct vertical component. Accept AWRT 13.

A1

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$V = \sqrt{13.01^2 + 16^2}$ = 20.6 m s^{-1}

dM1: Finding speed using their answer from part (a) and their vertical component.

A1:Correct final speed. Accept AWRT 20.6

dM1A1

5

(c) $\tan \alpha = \frac{13.01}{16}$ or $\sin \alpha = \frac{13.01}{20.6}$ or $\cos \alpha = \frac{16}{20.6}$

M1: Trig equation to find the angle with:

cos with 13 or 16 in the numerator and 20.6 in denominator

sin with 13 or 16 in the numerator and 20.6 in denominator

tan with 13 and 16 in any position

A1F: Correct equation.

M1A1F

$\alpha = 39.1°$

A1F: Correct angle. Accept AWRT 39°

A1F

Follow through incorrect answers to part (a) and (b), provided their speed from (b) is the resultant of two components.

3

[10]

Q4.

(a) $8 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on the vertical motion, with u = 0, s = ± 8 and a = ± 9.8.

A1: Correct equation.

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M1A1

$t = \sqrt{\frac{16}{9.8}}$ = 1.28 s

A1: Correct time. Allow 1.27 or AWRT 1.28.

A1

3

(b) $V \times \sqrt{\frac{16}{9.8}} = 20$

M1: Using 20 = speed × time.

A1: Correct equation.

M1A1

$V = 20\sqrt{\frac{9.8}{16}}$ = 15.7 m s^{-1}

A1: Correct speed. Accept 15.6 or 7 $\sqrt{5}$ or AWRT 15.6 or AWRT 15.7.

A1

3

(c) $v_y = 9.8 \times \sqrt{\frac{16}{9.8}}$ (= 12.52)

M1: Finding vertical component of velocity, with u = 0, a = ±9.8 and their time from part (a).
A1: Correct expression for velocity.

M1A1

$v = \sqrt{15.65^2 + 12.52^2}$ = 20.0 m s^{-1}

dM1: Finding the magnitude (with addition).

dM1

A1: Correct speed.
Accept 20 or 20.1 or AWRT 20.0.

A1

4

[10]

Q5.

(a) $\tan \alpha = \frac{6}{10}$

M1: Using tan with 10 and 5 or 6, OR sin or cos with $\sqrt{136}$ and 6 or 10, OR sin or cos with $\sqrt{125}$ and 5 or 10.

Note: $\sin \alpha = \frac{6}{\sqrt{136}}$ = and $\cos \alpha = \frac{10}{\sqrt{136}}$

M1

α = 31.0°

A1: Correct angle. Accept 30.9° or AWRT 31°

A1

2

(b) $8 \sin \alpha t + 4.9\, t^2 = 6$

M1: equation for the vertical motion containing ±6 or ±5, ± 4.9t^2 and ±8sinα or ±8cosα, where

α has a value related to their answer to part (a) (May be a negative angle).

M1

A1F: Correct terms.

A1F: Correct signs and terms Follow through angle from part (a).

A1FA1F

$4.9t^2 + 4.116t - 6 = 0$

A1: Correct equation rearranged equal to zero, but may be implied by subsequent working.

A1

$t = 0.76359$ or $t = -1.60$ s

dM1: Attempting to solve their quadratic equation. Only award method mark if method seen or correct answers obtained or –0.764 with +1.60.

dM1

$t = 0.764$

A1: Correct solution obtained. Accept 0.763 or AWRT 0.764.

A1

OR

$v = \sqrt{(8\sin 31.0°)^2 + 2 \times 9.8 \times 6} = 11.60$

M1: Use a constant acceleration equation $v^2 = u^2 + 2as$ to find v.

(M1)

A1F: Correct equation.

A1F: Correct v.

(A1FA1F)

$11.60 = 8\sin 31° + 9.8t$

dM1: Use of $v = u + at$ to find t

(dM1)

$t = \dfrac{11.60 - 8\sin 31°}{9.8} = 0.763$

A1: Correct equation.

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(A1)

A1: Correct t (0.763)

(A1)

6

(c) $d = 10 - 8\cos\alpha \times 0.764$

M1: Finding a horizontal distance
using 8cosα or 8sinα multiplied by their time from part (b).
dM1: For subtracting their distance from 10.

M1dM1

$= 10 - 5.238$

A1: Seeing AWRT 5.24 or 5.23 from 0.763.

A1

$= 4.76$ m

A1: Correct final answer. Accept AWRT 4.76.
Accept 4.77 from use of 0.763.

A1

4

[12]

Q6.

(a) $22.4\sin\theta - 2 \times 9.8 = 0$

M1: Use of $v = u + at$ vertically with
$u = 22.4\sin\theta$, $v = 0$, $t = 2$ and $a = \pm 9.8$.

A1: Correct equation. (May be in terms of g or contain 9.81..

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M1A1

$\sin\theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$

A1: Must see either

$22.4\sin\theta = 19.6$ or $\frac{19.6}{22.4}$

A1

OR

M1: Use of $s = ut + \frac{1}{2}at^2$ with $u = 22.4\sin\theta$, $s = 0$, $t = 4$ and $a = \pm 9.8$.

A1: Correct equation.

A1: must see $89.6\sin\theta = 78.4$ or $\frac{78.4}{89.6}$ OE

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3

(b) $h_{MAX} = 22.4 \times \frac{7}{8} \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with t = 2, u = 22.4 sin θ or 19.6 and a = ±9.8.

A1: Correct equation.

M1A1

= 19.6 m

A1: Correct height. AWRT 19.6

Note using g = 9.81 gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

3

(c) $\cos\theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04°$

B1: Correct value for cos θ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4$ m

M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times \frac{7}{8}t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct terms, but not necessarily signs.

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

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A1

t = 0.274 or t = 3.726

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:
M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

[14]

Q7.

(a) 22.4 sin θ – 2 × 9.8= 0

M1: Use of v = u + at vertically with u = 22.4 sin θ , v = 0, t = 2 and a = ±9.8.

A1: Correct equation. (May be in terms of g or contain 9.81..

M1A1

sin θ = $\frac{19.6}{22.4}$ = $\frac{7}{8}$ = 0.875

A1: Must see either

22.4 sin θ = 19.6 or $\frac{19.6}{22.4}$

AG

A1

Or

0 = 22.4 sin θ × 4 – $\frac{1}{2}$ × 9.8 × 4^2

M1: Use of s = ut + $\frac{1}{2}at^2$ with u = 22.4sinθ , s = 0, t = 4

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and a = ±9.8.
A1: Correct equation.

(M1A1)

$\sin\theta = \frac{4.9\times16}{22.4\times4} = 0.875$

A1: must see 89.6 sin θ = 78.4 or $\frac{78.4}{89.6}$ OE

(A1)

3

(b) $h_{MAX} = 22.4 \times \sin\theta \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with t = 2, u = 22.4 sin θ or 19.6 and a = ±9.8.
A1: Correct equation.

M1A1

= 19.6 m

A1: Correct height. AWRT 19.6

Note using g = 9.81 gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

Or

$0^2 = (22.4 \times \sin\theta)^2 + 2 \times (-9.8)h_{MAX}$

(M1A1)

h_{MAX} = 19.6m

(A1)

3

(c) $\cos\theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04°$

B1: Correct value for cosθ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4$ m

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M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times (\sin\theta)t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct terms, but not necessarily signs.

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

A1

$t = 0.274$ or $t = 3.726$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

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A1

Note: there are other methods which will lead to the correct time:

M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

(e) $v_{MIN} = 22.4 \times \cos\theta$

M1: Finding horizontal component with candidate's value for cos θ. Do not award if combined with a non-zero vertical component.

M1

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$= 10.8 \text{ m s}^{-1}$

A1: Correct speed. Accept 10.9 or 10.85.

A1

Or

$v_{min} = \frac{43.4}{4} = 10.9$ to 3sf

M1: range divided by time of flight

(M1)

A1: Correct speed. Accept 10.9 or 10.85.

(A1)

2

[16]

Q8.

(a) $x = ut \cos \alpha$

M1

$t = \frac{x}{u\cos\alpha}$

A1

$y = -\frac{1}{2}gt^2 + ut \sin \alpha$

Must have correct signs

M1

$y = -\frac{1}{2}g\left(\frac{x}{u\cos\alpha}\right)^2 + u\left(\frac{x}{u\cos\alpha}\right) \sin \alpha$

M1

$y = -\frac{gx^2}{2u^2\cos^2\alpha} + \frac{x\sin\alpha}{\cos\alpha}$

$y = -\frac{gx^2}{2u^2}(1 + \tan^2 \alpha) + x \tan \alpha$

A1

$k = -\frac{10(2k)^2}{2u^2}(1 + \tan^2 \alpha) + 2k \tan \alpha$

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M1

$u^2 = -20k(1 + \tan^2 \alpha) + 2u^2 \tan \alpha$

$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0$

AG

A1

7

(b) Pass through $P \Rightarrow$ Discriminant ≥ 0

$(-2u^2)^2 - 4(20k)(u^2 + 20k) \geq 0$

OE must be seen

M1A1

$4u^4 - 80ku^2 - 1600k^2 \geq 0$

$u^4 - 20ku^2 - 400k^2 \geq 0$

AG

A1

3

[10]

Q9.

(a) $12\sin 30° t - 4.9t^2 = -0.5$

$4.9t^2 - 12\sin 30° t - 0.5 = 0$

M1: Three term equation for vertical motion, with ±g, ±0.5 (or ±1 and ±1.5) and 12sin 30°t or 12cos30°t.

A1: Correct terms. (one must be equivalent to ±0.5)

A1: Correct signs.

M1A1A1

$t = 1.30281...$ *or* $-0.078323...$

dM1: Solving the quadratic to find t.

Must see use of quadratic equation formula or can be implied by seeing 1.303 or 1.302 or similar.

dM1

$t = 1.30$ seconds (to 3sf) AG

A1: Correct time from correct working. Must see more than 3 significant figures in candidate's working before the final answer or two correct solutions to the quadratic (eg 1.3 and –0.08).

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Accept 1.3

A1

OR

time up = 0.6122
time down = 0.6122 + 0.0783 = 0.6905
total time = 0.6122 + 0.6905 = 1.30 (to 3sf)

M1: Adding time up to time down having used a quadratic.

A1: 0.6122

dM1: Finding time down with a quadratic

A1: 0.6905

A1: Correct answer

Accept 1.3

(M1A1dM1A1A1)

OR

$-6.767 = 12 \sin 30° - gt$

M1:Forms an equation to find t having found v first

A1: Correct terms

A1: Correct signs

(M1A1A1)

$$t = \frac{12sin30° + 6.767}{g} = 1.30281 = 1.30 \text{ (to 3sf)}$$

dM1: Solving for t

A1: Correct time from correct working. Must see more than 3 significant figures in candidate's working before the final answer.

Accept 1.3

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(dM1A1)

5

(b) 12cos30° 1.303 = 13.5 m

M1: Finding horizontal displacement using 1.30 (or better) and 12cos30°.

Do not allow 12sin30°.

A1: Correct distance. AWRT 13.5.

M1A1

2

(c) v_y = 12sin30° – 9.8 1.3028 (= –6.767)

M1: Finding vertical component of velocity or velocity squared at impact.

Must include 12sin30° or 12cos30 and ±g

A1: Correct expression for vertical component.

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May have 1.3 or 1.30 instead of 1.3028.
(Accept +6.767 or similar)

M1A1

$v = \sqrt{(12cos30°)^2 + (-6.767)^2} = 12.4 \text{ m s}^{-1}$

dM1: Finding speed from two components. May use 6.74.

A1: Correct speed. Allow 12.3 or AWRT 12.4.

Note using g = 9.81 still gives 12.4.

dM1A1

4

(d) $\tan\theta = \dfrac{6.767}{12cos30°}$

M1: Trigonometric equation to find angle.

Can only be those shown opposite or described below. For tan, fraction can be inverted. For sin, 10.4 can be used instead of 6.767. For cos, 6.767 can be used instead of 10.4. Can use their values from part (c) (eg 6.74 or 6.77).

M1

θ = 33.1°

A1F: Correct angle. Accept AWRT 33°.

A1F

OR

$\sin\theta = \dfrac{6.767}{12.4}$

θ = 33.1°

OR

$\cos\theta = \dfrac{10.4}{12.4}$

θ = 33.1°

Follow though vertical component or final speed from part (c).

2

(e) The weight is the only force acting.
OR
No air resistance.

B1: Appropriate assumption.

B1

1

[14]

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Q10.

(a) 1000 = V × 4

M1: Equation for horizontal motion to find V.
Must not contain g. Could contain cos0° or equivalent.

M1

V = 250 ms^{-1}

A1: Correct V.

A1

2

(b) $(h =)\frac{1}{2} \times 9.8 \times 4^2$

M1: Vertical equation to find height with u = 0 and a = ±9.8.

M1

= 78.4 meters to 3sf

A1: Correct height. Accept –78.4

A1

2

(c) $(v_y =)9.8 \times 4 = 39.2 \text{ ms}^{-1}$

M1: Calculation of vertical component of velocity with u = 0 and a = ±9.8.
A1: Correct vertical component.

M1A1

or

$(v_y =)\sqrt{2 \times 9.8 \times 78.4} = 39.2 \text{ ms}^{-1}$

$(v_y =)\sqrt{250^2 + 39.2^2} = 253 \text{ ms}^{-1}$

dM1: Calculation of speed.
A1: Correct speed.

dM1A1

4

(d) $\tan\alpha = \frac{39.2}{250} \left(\text{or } \tan\alpha = \frac{250}{39.2}\right)$

M1: Using tan to find angle with opposite and adjacent sides. Can be inverted as shown in brackets.
A1F: Correct trig expression.

M1A1F

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α = 8.91°

A1: Correct angle.

A1

OR

$$\sin\alpha = \frac{39.2}{253}\left(\text{or } \sin\alpha = \frac{250}{253}\right)$$

M1: Using sin to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

α = 8.91°

A1: Correct angle.

(A1)

OR

$$\cos\alpha = \frac{250}{253(.055)}\left(\text{or } \cos\alpha = \frac{39.2}{253}\right)$$

M1: Using cos to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

α = 8.91°

A1: Correct angle. Accept 8.83° from this method.

(A1)

Note: Accept 8.98° from 253.1

Accept negative angles

Note: FT value of V from (a) and speed from (c) if needed. Do not FT 39.2 from (c) in place of 253.

Note: Accept energy methods if used correctly in part (c).

3

[11]

Q11.

(a) $x = 40\cos\theta.t$

M1

$$y = -\frac{1}{2}(10)t^2 + 40\ \sin\theta.t$$

M1 A1

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$$y = -\frac{1}{2}(10)(\frac{x}{40\ \cos\theta})^2 + 40\ \sin\theta.(\frac{x}{40\ \cos\theta})$$

Dependent on both M1s

m1

$$y = -\frac{x^2}{320\cos^2\theta} + x\tan\theta$$

$320\ y = -x^2(1 + \tan^2\theta) + 320x\tan\theta$

m1

$x^2\tan^2\theta - 320x\tan\theta + (x^2 + 320y) = 0$

Answer Given (Condone missing brackets)

A1

6

(b) (i) $150^2\tan^2\theta - 320(150)\tan\theta + (150^2 + 320 \times 8) = 0$

M1

$1125\tan^2\theta - 2400\tan\theta + 1253 = 0$

Correct quadratic

A1

$$\tan\ \theta = \frac{2400 \pm \sqrt{2400^2 - 4(1125)(1253)}}{2(1125)}$$

m1

$\tan\theta = 1.22$, 0.912

PI

A1F

$\theta = 50.7°$, $42.4°$

A1F

5

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(ii) $\theta = 42.4°$

For the smaller angle

B1F

$t = \dfrac{150}{40\cos\theta}$ and $\cos 42.4 > \cos 50.7$

OE

E1

2

[13]

Q12.

(a) $5 = \frac{1}{2} \times 9.8t^2$

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M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$t = \sqrt{\frac{5}{4.9}} = 1.01$ s **AG**

A1: Correct time from correct working.

Must see square root or t^2 = 1.02 OE

Note: If g = 9.81 is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed × time OE

M1

$V = 15\sqrt{\frac{4.9}{5}} = 14.8$

A1: Correct speed.
Accept AWRT 14.8 or 14.9.

Note: If g = 9.81 is used for the first time deduct one mark. Should get 14.9 m s^{-1} from g = 9.81.

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A1

2

(c) $v_v = \pm 9.8 \times \sqrt{\frac{5}{4.9}}$ (= ± 9.899)

M1: Calculating vertical component of velocity.
A1: Correct value. Accept 9.9 or similar

M1A1

or

$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$

$v = \sqrt{9.899^2 + 14.8^2}$ = 17.8 m s^{-1}

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

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dM1

A1: Correct speed.
Accept AWRT 17.8 or AWRT 17.9.
Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.9 m s^{-1} from g = 9.81

A1F

4

(d) $\tan\alpha = \frac{9.899}{14.8}$ or $\frac{14.8}{9.899}$

$\alpha = 34°$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

3

$\sin\alpha = \frac{9.899}{17.8}$ or $\frac{14.8}{17.8}$

$\alpha = 34°$

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

(M1)(A1F)(A1F)

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$\cos\alpha = \frac{14.8}{17.8}$ or $\frac{9.899}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

[12]

Q13.

(a) $5 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

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A1: Correct equation

A1

$t = \sqrt{\frac{5}{4.9}} = 1.01\text{ s}$ **AG**

A1: Correct time from correct working.
Must see square root or $t^2 = 1.02$ OE
Note: If $g = 9.81$ is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed × time OE

M1

$V = 15\sqrt{\frac{4.9}{5}} = 14.8$

A1: Correct speed.
Accept AWRT 14.8 or 14.9.
Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 14.9 m s^{-1} from $g = 9.81$.

A1

2

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(c) $v_v = \pm 9.8 \times \sqrt{\frac{5}{4.9}} (= \pm 9.899)$

M1: Calculating vertical component of velocity.
A1: Correct value. Accept 9.9 or similar

M1A1

or

$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$

$v = \sqrt{9.899^2 + 14.8^2} = 17.8\text{ m s}^{-1}$

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

A1: Correct speed.
Accept AWRT 17.8 or AWRT 17.9.

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A1F

Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.9 m s^{-1} from g = 9.81

4

(d) $\tan\alpha = \frac{9.899}{14.8}$ or $\frac{14.8}{9.899}$

$\alpha = 34°$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

3

$\sin\alpha = \frac{9.899}{17.8}$ or $\frac{14.8}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

$\cos\alpha = \frac{14.8}{17.8}$ or $\frac{9.899}{17.8}$

$\alpha = 34°$

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(M1)(A1F)(A1F)

(e) Particle

B1: Particle assumption

B1

Experiences no air resistance **or** no wind **or** only gravity **or** no other forces acting **or** no spin.

B1: Other assumption.
Ignore any other assumptions.

B1

2

[14]

EXAM PAPERS PRACTICE

Q14.

(a) $14.7 \sin \alpha - 9.8t \ (= 0)$

M1: Equation for vertical velocity being zero at highest point. Must have sin α with ± 9.8.
A1: Correct equation.

M1A1

$$t = \frac{14.7 \sin \alpha}{9.8} = \frac{3 \sin \alpha}{2} \quad \textbf{AG}$$

A1: Correct result from correct working.

A1

OR

$$14.7 \sin \alpha T - 4.9T^2 (= 0)$$
$$T = \frac{14.7 \sin \alpha}{4.9} = 3 \sin \alpha$$
$$t = \frac{3 \sin \alpha}{2}$$

All marks awarded for last line, from correct working.

(M1)(A1)(A1)

3

(b) (i) $$7 = 14.7 \sin \alpha \left(\frac{3 \sin \alpha}{2}\right) - 4.9\left(\frac{3 \sin \alpha}{2}\right)^3$$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with ± g or equivalent.

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M1

A1: Correct expression.

A1

$$7 = 11.025 \sin^2 \alpha$$

dM1: Simplified expression with sin² α.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

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$$\alpha = \sin^{-1}\left(\sqrt{\frac{7}{11.025}}\right) = 52.8°$$

A1: Correct angle.
Accept 52.7°, 52.9°.

A1

OR

$$0^2 = (14.7\sin\alpha)^2 + 2\times(-9.8)\times 7$$

(M1)(A1)

$$\sin^2\alpha = \frac{2\times 9.8\times 7}{14.7^2}$$

(dM1)

$$\alpha = 52.8°$$

(dM1)(A1)

5

(ii) OA = 14.7 cos 52.8° × 3 sin 52.8°

B1: Use of 3sin α with their α.
M1: Finding horizontal displacement.
including 14.7 cos α with 3 sin α or $\frac{3\sin\alpha}{2}$.

B1M1

OA = 21.2 m

A1: Correct distance. Accept 21.3 m.

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A1

3

Use of g = 9.81:
(a) M1A1A0
(b) (i) 52.8° or 52.9° M1A1dM1dM1A1
(b) (ii) 21.2 B1M1A1

[11]

Q15.

(a) $14.7\sin\alpha - 9.8t (= 0)$

M1: Equation for vertical velocity being zero at highest point. Must have sin α with ± 9.8.
A1: Correct equation.

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M1A1

$t = \dfrac{14.7\sin\alpha}{9.8} = \dfrac{3\sin\alpha}{2}$ **AG**

A1: Correct result from correct working.

A1

OR

$14.7\sin\alpha T - 4.9T^2 (= 0)$

$T = \dfrac{14.7\sin\alpha}{4.9} = 3\sin\alpha$

$t = \dfrac{3\sin\alpha}{2}$

All marks awarded for last line, from correct working.

(M1)(A1)(A1)

3

(b) (i) $7 = 14.7\sin\alpha\left(\dfrac{3\sin\alpha}{2}\right) - 4.9\left(\dfrac{3\sin\alpha}{2}\right)^2$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with ±g or equivalent.

M1

A1: Correct expression.

A1

$7 = 11.025\sin^2\alpha$

dM1: Simplified expression with $\sin^2\alpha$.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$\alpha = \sin^{-1}\left(\sqrt{\dfrac{7}{11.025}}\right) = 52.8°$

A1: Correct angle.
Accept 52.7°, 52.9°.

A1

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OR

$0^2 = (14.7\sin\alpha)^2 + 2\times(-9.8)\times 7$

(M1)(A1)

$\sin^2\alpha = \dfrac{2\times 9.8\times 7}{14.7^2}$

(dM1)

$\alpha = 52.8°$

(dM1)(A1) **5**

(ii) OA = 14.7 cos 52.8° × 3 sin 52.8°

B1: Use of 3sin α with their α.
M1: Finding horizontal displacement.

including 14.7 cos α with 3 sin α or $\dfrac{3\sin\alpha}{2}$.

B1M1

OA = 21.2 m

A1: Correct distance. Accept 21.3 m.

A1 **3**

(c) Ball is a particle/No spin.

B1: Particle assumption.

B1

No air resistance/No wind/Constant acceleration of 9.8/Only force is weight.

B1: Air resistance assumption.

B1 **2**

[13]

Use of g = 9.81:

(a)	*M1A1A0*
(b) (i) 52.8° or 52.9°	*M1A1dM1dM1A1*
(b) (ii) 21.2	*B1M1A1*

Q16.

(a) (i) $x = 80\cos\theta . t$

B1

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$t = \dfrac{x}{80\cos\theta}$

B1

$y = 80\sin\theta . t - \dfrac{1}{2}gt^2$

B1

$y = 80\sin\theta \dfrac{x}{80\cos\theta} - \dfrac{1}{2}g(\dfrac{x}{80\cos\theta})^2$

M1

$y = x\tan\theta - \dfrac{gx^2}{12800}(1+\tan^2\theta)$

Answer given

A1 5

(ii) $-20 = 400\tan\theta - \dfrac{9.8 \times 400^2}{12800}(1+\tan^2\theta)$

Condone + 20

M1

$122.5\tan^2\theta - 400\tan\theta + 102.5 = 0$

$49\tan^2\theta - 160\tan\theta + 41 = 0$

Answer given

A1 2

(b) (i) $\tan\theta = \dfrac{160 \pm \sqrt{25600 - 4(49)(41)}}{2 \times 49}$

M1

= 2.9850, 0.2803

PI

A1

$\theta = 71.5°, 15.7°$

A1F 3

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(ii) For the shortest time

$400 = 80 \cos 15.7°.t$

M1

$t = 5.19$

A1F

2

(c) • The projectile is a particle
• The air resistance is negligible

E1

1

[13]

Q17.

(a) $0^2 = (30 \sin 35°)^2 + 2 \times (-9.8)s$

M1: Equation to fine the max height, with = 0

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$$s = \frac{(30 \sin 35°)^2}{2 \times 9.8} = 15.1$$ m

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35°)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$$v_y = \sqrt{28^2 - (30 \cos 35°)^2}$$

M1: Solving equation

M1

$= 13.4198 = 13.4 \text{ m s}^{-1}$ **AG**

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A1: Correct speed from correct working.

A1

4

(c) $\tan\theta = \dfrac{13.4}{30\cos 35°}$

M1: Expression to find the angle.

M1

$\theta = 28.6°$

A1: Correct angle.

A1

2

[10]

Q18.

If candidates have already used g = 9.81 do not penalise again on this question.

(a) $0^2 = (28\sin 50°)^2 + 2\times(-9.8)s$

M1: Equation to find the max height, with v = 0, u = 28 sin 50° or u = 28 cos 50° and – 9.8 or –g

M1

A1: Correct equation

A1

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$s = \dfrac{(28\sin 50°)^2}{2\times 9.8} = 23.5\text{ m}$

dM1: Solving for the height

dM1

A1: Correct height. Awrt 23.5
Note: If using a memorised formula, either 4 marks if final answer correct, 3 marks if substituted correctly but evaluated incorrectly, otherwise zero.

A1

OR

$0 = 28\sin 50° - 9.8t$

M1: Equation to find time to the max height, with v = 0, u = 28 sin 50° or

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u = 28 cos 50° and – 9.8 or – g

(M1)

$$t = \frac{28\sin 50°}{9.8} = 2.1887$$

A1: Correct time

(A1)

$$s = 28\sin 50° \times 2.1887 - 4.9 \times 2.1887^2 = 23.5$$

dM1: Finding the height with their time and u = 28 sin 50° or u = 28 cos 50° and –4.9 or –g/2

(dM1)

A1: Correct height. Awrt 23.5

(A1)

4

(b) 2 = 28 sin 50° t – 4.9t^2

M1: Quadratic equation in t with a ± 2, u = 28 sin 50° or u = 28 cos 50° and –4.9 or –g/2.

M1

A1: Correct terms

A1

A1: Correct signs for equation

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A1

0 = 4.9t^2 – 28 sin 50° t + 2
t = 0.0953 or t = 4.282

dM1: Solving the quadratic equation

dM1

t = 4.282 = 4.28 s (to 3 sf) **AG**

A1: Correct larger time selected from two values.

A1

OR

M1: Calculation of two times, which sum or differ to give the time of flight.

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(M1)

$0 = 28\sin 50° - 9.8t$

$t = \dfrac{28\sin 50°}{9.8} = 2.1887$

A1: Correct time by equation for zero vertical component of velocity or maximum height.

(A1)

OR

$23.5 = 28\sin 50° t - 4.9t^2$

$t = 2.1887$

$21.5 = 4.9t^2$

dM1: Correct expression for time to fall.

(dM1)

$t = \sqrt{\dfrac{21.5}{4.9}} = 2.0947$

A1: Correct time.

(A1)

2.1887 + 2.0947 = 4.2834 = 4.28 (to 3 sf) **AG**

A1: Correct time. Accept 4.29 if their answer rounds to 4.29.

(A1)

5

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(c) $v_x = 28\cos 50° (= 18.00\ \text{m s}^{-1})$

B1: Horizontal component, need not be evaluated.

B1

$v_y = 28\sin 50° - 9.8 \times 4.282 = -20.51\ \text{m s}^{-1}$

M1: Equation for vertical component with 28 sin 50° (or 28 cos 50° if sin 50° used for horizontal component), −9.8 and awrt 4.28.

M1

A1: Correct vertical component. Awrt ± 20.5

A1

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$v = \sqrt{18.00^2 + 20.51^2} = 27.3 \text{ m s}^{-1}$

dM1: Finding speed with a + sign inside the square root.

dM1

A1F: Correct speed. Awrt 27.3.

A1F

5

Intermediate values can be implied by final answer.

[14]

Q19.

(a) $h = \frac{1}{2} \times 9.8 \times 0.6^2 = 1.76$ m AG

M1: Equation to find h

M1

A1: Correct value from correct working

A1

2

(b) $x = 18 \times 0.6 = 10.8$ m

M1: Calculating range

M1

A1: Correct value

A1

2

(c) $v_y = 9.8 \times 0.6 = 5.88 \text{ m s}^{-1}$

B1: Correct vertical

$v = \sqrt{5.88^2 + 18^2} = 18.9 \text{ m s}^{-1}$

M1: Calculating speed

M1

A1: Correct speed

A1

$\theta = \tan^{-1}\left(\frac{5.88}{18}\right) = 18.1°$

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M1: Calculating angle using tan

M1

A1: Correct angle

A1

18.9 m s^{-1} at 18.1° below the horizontal.

B1: States below horizontal

B1

6

[10]

Q20.

(a) 20 sin 50° t – 4.9 t^2 = 0

M1: Equation to find time, with y = 0,
u = 20 sin 50° or u = 20 cos 50° and ± 9.8 or ± g.
A1: Correct equation

M1A1

$t = \dfrac{20 \sin 50°}{4.9}$ or 3.126 ... = 3.13 s **AG**

dM1: Solving for t.
A1: Correct time from correct working.
Must see division by 4.9 oe or more than 3 sf
Verification methods can only gain first 2 marks

Special case $t = \dfrac{15.3}{4.9} = 3.12$ or 3.13 *scores*
M1A1dM1A0

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dM1A1

OR

$0 = 20 \sin 50° - 9.8t$

$t = \dfrac{20 \sin 50°}{9.8} = 1.563$

$T = 2 \times 1.563$

M2: doubling time to max height (could use cos instead of sin) but must use ± 9.8 or ± g.

(M2)

= 3.13

A2: Correct time from correct working.

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Don't penalise use of g = 9.81 if already done earlier on script. Would obtain time as 3.12 seconds.
Note: If using a memorised formula either 4 marks if final answer correct, 3 marks if substituted correctly, otherwise zero.

Special case
T = 2 × 1.56 = 3.12 or 3.13 scores M2A1

(A2)

4

(b) $PQ = 20 \cos 50° \times 3.127 = 40.2$ m

M1: Calculation of range, could use sin instead of cos.

M1

A1: Correct range
Accept 40.1

A1

2

(c) No change because a greater mass would not change the acceleration. OR Mass is not used in the equations.

B1: No change

B1

B1: Explanation following a correct statement.

B1

2

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(d) $0 = (20 \sin 50°)^2 + 2 \times (-9.8)s$

M1: Equation to find height, with u = 20 sin 50° or u = 20 cos 50° and ± 9.8 or ± g (and t between 1.56 and 1.57 if method 2 used).
A1: Correct equation

M1A1

$$s = \frac{(20 \sin 50°)^2}{2 \times 9.8} = 12.0 \text{ m}$$

A1: Correct height. Accept 12 or 11.9 or AWRT 12.0

A1

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OR

$t = \frac{3.13}{2} = 1.565$

In method 2, no marks awarded for just finding t.

$h = 20 \sin 50° \times 1.565 - 4.9 \times 1.565^2$

(M1)(A1)

= 12.0

(A1)

Don't penalise use of g = 9.81 if already done earlier on script. Should still get 12. Note: If using a memorised formula either 3 marks if final answer correct, 2 marks if substituted correctly, otherwise zero.

3

(e) 20 m s^{-1} at 50° below the horizontal.

B1: Speed AWRT 20

B1

B1: Direction AWRT 50°. Must indicate below, or down. Could be implied by a diagram.

B1

2

[13]

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Q21.

(a) $x = 2t$

M1

$y = -\frac{1}{2}gt^2 + 10t$

M1

$t = \frac{x}{2}$

$y = -\frac{1}{2}g\left(\frac{x}{2}\right)^2 + 10\left(\frac{x}{2}\right)$

m1

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$y = -\frac{g}{8}x^2 + 5x$

AG

A1

4

(b) $1 = -\frac{g}{8}x^2 + 5x$

M1

$gx^2 - 40x + 8 = 0$

$x = \frac{40 \pm \sqrt{(-40)^2 - 4 \times 8g}}{2g}$

M1

x = 3.871, 0.211

A1 for both answers

A1

Distance = 3.66m

A1

4

(c) $t = \frac{3.66}{2}$

M1

$t = 1.83$ sec

A1

2

[10]

Q22.

(a) $-3 = 70\sin 10°t - 4.9t^2$

Vertical equation to find t containing a 3

M1

Correct RHS of equation

A1

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$4.9t^2 - 70\sin 10°t - 3 = 0$

Correct LHS of equation

A1

$$t = \frac{70\sin 10° \pm \sqrt{(70\sin 10°)^2 - 4 \times 4.9 \times (-3)}}{2 \times 4.9}$$

Solving for t

dM1

$t = -0.226$ or 2.71

Selecting the positive t

dM1

$t = 2.71$

Correct time from correct working

A1

6

(b) $s = 70\cos 10° \times 2.71 = 187$

Finding horizontal distance

M1

Correct distance

A1

2

(c) $v_x = 70\cos 10° = 68.9$

Calculating horizontal component of velocity

M1

Correct value

A1

2

(d) $v_y = 70\sin 10° - 9.8 \times 2.71 = -14.4$

Calculating vertical component of velocity at t = 2.71

M1

Correct value

A1

$v = \sqrt{68.9^2 + 14.4^2} = 70.4$

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Finding speed

dM1

Correct speed

A1

4

[14]

Q23.

(a) It is a particle /No air resistance / lift forces act on the ball.

Particle

B1

Other acceptable assumption
Deduct one mark for each additional incorrect assumption.

B1

2

(b) $V\sin 40° t - \frac{1}{2} \times 9.8t^2 = 0$

Vertical equation to find t.

M1

Correct equation
(Equals zero may be implied)

A1

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$t = \frac{V\sin 40°}{4.9}$

Solving for t

dM1

Correct t

A1

$s = V\cos 40° \times \frac{V\sin 40°}{4.9}$ **AG**

$= \frac{V^2\cos 40°\sin 40°}{4.9}$

Finding range with their t

M1

Correct range from correct working

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SC Quoting the formula for the range 2 marks.

A1

6

(c) $$76 < \frac{V^2 \cos 40° \sin 40°}{4.9} < 82$$

$$\sqrt{\frac{76 \times 4.9}{\cos 40° \sin 40°}} < V < \sqrt{\frac{82 \times 4.9}{\cos 40° \sin 40°}}$$

$27.5 < V < 28.6$

An equation to find one value of V.

M1

Correct value for V

A1

Other value of V correct

A1

Correct range of values
Accept 27.5 – 28.6 but not 28.6-27.5
For using values close to 76 and 82 deduct one mark.

A1

4

[12]

Q24.

(a) $0 = 16 \sin \theta \times 2 - 4.9 \times 2^2 = 0$ **AG**

Equation to find θ based on height being zero.

M1

Correct equation

A1

$$\sin \theta = \frac{19.6}{32}$$

$\theta = 37.8°$

Correct angle from correct working

A1

3

(b) $2 = 16 \sin 37.8t - 4.9t^2$

Equation for height of 2

M1

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Correct equation

A1

$4.9t^2 - 16\sin 37.8t + 2 = 0$

$$t = \frac{16\sin 37.8 \pm \sqrt{(16\sin 37.8)^2 - 4\times 2\times 4.9}}{2\times 4.9}$$

Solving quadratic equation

m1

$t = 1.77$ or $t = 0.23$

Correct solutions

A1

Greater than 2 metres for 1.54 seconds

Difference between solutions

A1

5

[8]

Q25.

(a) (i) $0 = 40\sin 35°t - 4.9t^2$

Equation to find time of flight with 40,

sin/cos35° and –4.9 or $-\frac{g}{2}$

M1

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Correct equation

A1

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$$t = \frac{40\sin 35°}{4.9} = 4.68 \text{ s}$$ **AG**

Solving for t

m1

Correct time from correct working
Note: candidates must have a method for the complete time of flight before any marks can be awarded.
Condone the use of a formula for the time of flight.

A1

4

(ii) $AB = 40\cos 35° \times 4.682 = 153$ m

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Calculating the range using 40, cos/sin35°
and 4.68 and acceleration zero.

M1

Correct range
Accept AWRT 153

A1

2

(b) $-1 = 40\sin 35° t - 4.9t^2$

$4.9t^2 - 40\sin 35° t - 1 = 0$

Equation to find time of flight with a ±1,

40, sin/cos35° and –4.9 or $-\frac{g}{2}$

M1

Correct terms

A1

Correct signs

A1

$$t = \frac{40\sin 35° \pm \sqrt{(40\sin 35°)^2 - 4 \times 4.9 \times (-1)}}{2 \times 4.9}$$

Solving quadratic equation

m1

$t = 4.73$ or $t = -0.0432$

Accept AWRT 4.73 or 4.72

A1

$t = 4.73$

Rejection of negative solution indicated
(Only 4.73 or 4.72 given award 5/6 marks)

A1

Alternative methods based on finding two times.

For example,

$t = 4.682 + 0.044 = 4.73$

$t = 2.341 + 2.384 = 4.73$

Addition of two times

(M1)

Use of AWRT 4.68 or AWRT 2.34

(A1)

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Calculation of time for 'second' part

(m1)

Correct expression for time for 'second' part

(A1)

Correct time (Allow AWRT 0.04 or AWRT 2.38)

(A1)

Correct total time
Accept 4.72

(A1)

6

[12]

Q26.

(a) $5 = 10 \cos \alpha . t$

M1

$$t = \frac{5}{10 \cos \alpha}$$

A1

$$1 = -\frac{1}{2}(9.8)t^2 + 10 \sin \alpha t$$

M1A1

$$1 = -\frac{1}{2}(9.8)\frac{25}{100 \cos^2 \alpha} + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

Dependent on both M1s

M1

$$1 = -\frac{1}{2}(9.8)\frac{25}{100}(1 + \tan^2 \alpha) + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

A1

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

Answer given

A1

7

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(b) $\tan\alpha = \dfrac{200 \pm \sqrt{40000 - 4(49)(89)}}{2 \times 49}$

M1

= 3.57, 0.508

AWRT

A1

$\alpha = 74.4°, 26.9°$

A1F

3

(c) (i) 10 cos 26.9° = 8.92 (or 8.91) > 8

$\Rightarrow$ The can will be knocked off the wall

Both values checked

M1

10 cos 74.4° = 2.69 < 8

Acc. of both results
Correct conclusions

A1F

$\Rightarrow$ The can will not be knocked off the wall

E1

ALTERNATIVE
The can will be knocked off the wall if
10 cos α > 8
cos α > 0.8
α < 36.9° *M1A1*
So, for α = 26.9° the can will be knocked off
and for α = 74.4°, the can will not be knocked off *E1*

3

(ii) $x = ut$

$t = \dfrac{5}{10\cos 26.9°}$

$v = 10\sin 26.9° - 9.8\left(\dfrac{5}{10\cos 26.9°}\right)$

Any correct use of equations

M1

$v = -0.970$

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A1F

$\tan\theta = \frac{-0.970}{8.92}$

M1

$\theta = -6.2°$

At an angle of depression of 6.2°

AWRT 6°

A1F

4

[17]

Q27.

(a) $6 = 20 \sin 60° t - 4.9t^2$

Vertical equation including a 6

M1

Correct terms

A1

$4.9t^2 - 20 \sin 60° t + 6 = 0$

Correct signs

A1

$$t = \frac{20\sin 60° \pm \sqrt{(20\sin 60°)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic equation

dM1

$t = 0.389$ or $t = 3.15$

$t = 0.389$ s

Correct solution selected from correct working; AG

A1

5

(b) $x = 20 \cos 60° \times 0.389 = 3.89$ m

Finding distance. Correct distance

M1A1

2

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(c) $v_x = 20\cos 60° = 10$

Correct horizontal component

B1

$v_y = 20\sin 60° - 9.8 \times 0.389 = 13.5$

Finding vertical component of velocity

M1

Correct vertical component

A1

$v = \sqrt{10^2 + 13.5^2} = 16.8\ \text{m s}^{-1}$

Finding magnitude. Correct magnitude

dM1A1

5

[12]

Q28.

(a) $0^2 = (50\sin 40°)^2 + 2 \times (-9.8)h$

Equation for h with v = 0 and a component of velocity. Correct equation

M1A1

$$h = \frac{(50\sin 40°)^2}{2 \times 9.8} = 52.7$$

Solving for h

dM1

Correct h

A1

Alt

$0 = 50\sin 40° - 9.8t$

Equation for t with v = 0 and a component of velocity

(M1)

$$t = \frac{50\sin 40°}{9.8} = 3.280$$

Correct t

(A1)

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$h = 50 \sin 40° \times 3.280 - \frac{1}{2} \times 9.8 \times 3.280^2$

Expression for h with a component of velocity

(dM1)

= 52.7
ALLOW 52.6

Correct h

(A1)

4

(b) $6 = 50 \sin 40° t - 4.9t^2$

Forming a quadratic in t. Correct terms with any signs

M1A1

$0 = 4.9t^2 - 50 \sin 40° t + 6$

Correct equation

A1

$$t = \frac{50 \sin 40° \pm \sqrt{(50 \sin 40°)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic

dM1

= 0.192 or 6.37
$t = 6.37$

Correct solution selected

A2

Alt

$46.7 = 4.9 t_1^2$

Finding two times

(M1)

Equation for time to go down

(dM1)

$t_1 = 3.087$

Correct time

(A1)

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$t_2 = 3.280$

Time to go up

(A1)

$t = 3.087 + 3.280 = 6.37$

Correct total

(A2)

6

[10]

Q29.

(a) $2.45 = \frac{1}{2} \times 9.8t^2$

Equation for time to ground

M1

Correct equation

A1

$t = \sqrt{\frac{2.45}{4.9}} = 0.707 \text{ seconds (to 3 sf)}$

AG Correct time from correct working

A1

3

(b) $s = 20 \times 0.707 = 14.1$ m (to 3sf)

Using 20 × time from part (a)

M1

Correct distance

A1

2

(c) $v_y = 0.707 \times 9.8 = 6.929$

Finding vertical component of velocity

M1

Correct vertical component

A1

$\theta = \tan^{-1}\left(\frac{6.929}{20}\right) = 19.1°$

Using tan to find the angle

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dM1

Correct angle

A1

4

[9]

Q30.

(a) A particle or no spin

First assumption

B1

No air resistance or no wind or only gravity acting

Second assumption

B1

If more than 2 assumptions given, subtract one mark for each incorrect additional assumption

2

(b) $0 = 25 \sin 40° t - 4.9t^2$

Equation for time of flight

M1

Correct equation

A1

$0 = t\,(25 \sin 40° - 4.9t)$

Solving for t

dM1

$t = 0$ or $t = \dfrac{25 \sin 40°}{4.9}$

Time of flight = 3.28 s

AG Correct final answer from correct working (Verification method M1A1M1A0)

A1

4

(c) $s = 3.28 \times 25 \cos 40° = 62.8$ m

Finding range

M1

Correct range

A1

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2

(d) 25 m s^{-1} at 40° **below** the horizontal

Speed

B1

Direction

B1

2

(e) v_{min} = 25 cos40° = 19.2 m s^{-1}

OR

$$v_{min} = \frac{62.807}{3.2795} = 19.2\,\text{m s}^{-1}$$

Horizontal component of velocity

M1

Correct speed
Accept 19.1 m s^{-1}

A1

2

[12]

Q31.

(a) $$y = ut\sin\alpha - \frac{1}{2}gt^2$$

M1A1

$$x = ut\cos\alpha$$

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M1

$$t = \frac{x}{u\cos\alpha}$$

A1

$$y = u\left(\frac{x}{u\cos\alpha}\right)\sin\alpha - \frac{1}{2}g\left(\frac{x}{u\cos\alpha}\right)^2$$

M1

$$y = x\tan\alpha - \frac{gx^2}{u^2\cos^2\alpha}$$

$$y = x\tan\alpha - \frac{gx^2}{2u^2}(1+\tan^2\alpha)$$

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Answer given

A1 6

(b) (i) $1 = R\tan\alpha - \dfrac{10R^2}{2u^2}\left(1 + \tan^2\alpha\right)$

M1

$5R^2\tan^2\alpha - u^2 R\tan\alpha + 5R^2 + u^2 = 0$

Answer given

A1 2

(ii) For real solutions of the quadratic:

$u^4R^2 - 20R^2(5R^2 + u^2) \geq 0$

M1

$R^2 \leq \dfrac{u^4 - 20u^2}{100}$

$R^2 \leq \dfrac{u^2(u^2 - 20)}{100}$

Answer given

A1 2

(iii) $5^2 \leq \dfrac{u^2(u^2 - 20)}{100}$

$u^4 - 20u^2 - 2500 \geq 0$

Condone equation

M1

$u_{\text{min}}{}^2 = 61.0 \quad (\text{or } 10 + \sqrt{2600})$

A1

$u_{\text{min}} = 7.81$

3 sf required

A1F 3

[13]

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Q32.

(a) $s = ut + \frac{1}{2}at^2$

$0 = 2\frac{1}{2}ut - \frac{1}{2}gt^2$

M1

Full method required for time
(equations of motion, or standard result)

A1

$0 = t\left(2\frac{1}{2}u - \frac{1}{2}gt\right)$

m1

$t = \frac{5u}{g}$

(if g *= 9.8 used, lose last A1)*

A1

4

(b) $OA = 6u \times \frac{5u}{g}$

M1

$= \frac{30u^2}{g}$

CAO

A1

2

(c) speed $^2 = (6u)^2 + \left(2\frac{1}{2}u\right)^2$

M1

speed = $6\frac{1}{2}u$

CAO

A1

2

(d) Least speed, at top, $= 6u$

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B1

1

[9]

Q33.

(a) $57 = 24\cos 40^\circ \times t$

component attempted and acceleration = 0

M1

all correct

A1

$t = 3.10$ sec

CAO

A1

3

(b) $h = 24\sin 40^\circ \times 3.1 - \frac{1}{2} \times 9.8 \times 3.1^2$

component attempted & acceleration = 9.8

M1

all correct

A1

$h = 0.734$ m

ft one slip e.g. + 9.8 used
Accept 2 SF answer, AWRT 0.71-0.74

A1F

3

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(c) (i) horizontal, $u = 24\cos 40^\circ = 18.39$ ms^{-1}

Seen anywhere in (c), accept 18.4

B1

vertical, $v = 24\sin 40^\circ - 9.8 \times 3.1$

component attempted & acceleration = 9.8

M1

$v = -14.95$ ms^{-1}

(accept -15.0)

A1

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$V = \sqrt{(18.39^2 + (-14.95)^2)}$

Use of candidate's u and 'new' v

M1

$V = 23.7 \text{ms}^{-1}$

ft use of candidate's u and v

A1F

5

(ii) $\tan\theta = \frac{14.95}{18.39}$

Use of candidate's u and v (new)
Accept inverted ratio

M1

$\theta = 39.1° \text{ or } 39.2°, 140.8°, 140.9° \text{ accept } \pm$

ft use of candidate's u and v and V

A1

2

[13]

Q34.

(a) $y = -\frac{1}{2}gt^2 + 20\sin 30.t$

M1A1

$x = 20\cos 30.t$

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M1

$t = \frac{x}{20\cos 30}$

A1

$y = -\frac{1}{2}g\frac{x^2}{400\cos^2 30} + 20\sin 30\frac{x}{20\cos 30}$

M1

$y = x\tan 30 - \frac{gx^2}{800\cos^2 30°}$

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AG

A1

6

(b) $2.5 = x\tan 30 - \frac{9.8x^2}{800\cos^2 30}$

$9.8x^2 - 346x + 1500 = 0$

substituting and tidying up

M1A1

$x = \frac{346 \pm \sqrt{119716 - 58800}}{19.6}$

M1

=30.3 (or 30.2) & 5.06 (or 5.05)
answer: 30.3m (or 30.2m)
at least 3 s.f. required

A1F

4

(c) no air resistance,

B1

the ball is a particle

B1

2

etc.

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[12]

Q35.

(a) (i) $30 \sin 60°t - 4.9t^2 = 0$

$t = \frac{30\sin 60°}{4.9} = 5.302$ AG

Expression for height equal to zero

M1

Correct expression

A1

AG; correct conclusion from correct working

A1

3

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(ii) $s = 30 \cos 60° \times 5.302 = 79.5$ m

Use of horizontal component of velocity to find the range

M1

Correct range

A1

2

(b) $30 \sin 60°t - 4.9t^2 = 5$

$4.9t^2 - 30 \sin 60°t + 5 = 0$

$t = 0.200$ or 5.102

$s = 30 \cos 60° \times 5.102 = 76.5$ m

Expression for height equal to 5

M1

Correct LHS

A1

Correct RHS

A1

Solving a quadratic equation

m1

Correct solutions

A1

Correct distance

A1

6

[11]

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